

An application of the method of finite Markov chain imbedding to runs tests

W.Y. Wendy Lou

*Department of Biomathematical Sciences, Box 1023, Mount Sinai School of Medicine, One Gustave L. Levy Place,
New York, NY 10029-6574, USA*

Received September 1995; revised November 1995

Abstract

The method of finite Markov chain imbedding developed by Fu and Koutras (1994) has become a popular and useful tool for studying runs and patterns-related problems. In this article, their approach is used as an alternative for obtaining the exact conditional distribution of the success runs statistic given the number of successes in a sequence of n i.i.d. Bernoulli trials, which, in the past, was computed mainly from traditional combinatorics.

Keywords: Bernoulli trials; Transition probability matrix; Success runs

1. Introduction

In the past, runs and runs-related statistics were studied mainly as combinatorial problems (e.g. Mood, 1940; Wald and Wolfowitz, 1940; David, 1947; Hirano, 1986; Godbole, 1990). Because of the complexity of the combinatorial approach, many exact distributions of runs statistics remain unknown, especially for non-i.i.d. cases. Recently, Fu and Koutras (1994) and Fu (1996) systematically developed, away from the lines of combinatorics, formulae for the exact distributions of certain runs and patterns associated with Bernoulli and multi-state trials. The basis of their approach is to imbed the problem, or say the random variable of interest, into a finite Markov chain. The exact distribution of the random variable can then be expressed simply in terms of the transition probability matrices of the imbedded Markov chain. This approach is an extension of earlier developments by Fu (1986) for studying certain reliability problems.

In this article, the exact conditional distribution of the success runs statistic given the number of successes in a sequence of n i.i.d. Bernoulli trials is derived by applying the method of finite Markov chain imbedding directly, as opposed to the indirect approach developed by Lou (1996). The main focus throughout will be to demonstrate the intuitive simplicity of this approach. An example is also given to show that finite Markov chain imbedding is not necessarily unique. The finite Markov chain imbedding technique is briefly reviewed below before proceeding to the main result.

For a given positive integer n , let $\Gamma_n = \{0, 1, \dots, n\}$ be a finite index set, and $\Omega = \{a_1, \dots, a_k\}$ be a finite state space with k states. A non-negative integer random variable X_n , defined from a sequence of n Bernoulli

or multi-state trials, is “finite Markov chain imbeddable” if (i) there exists a finite Markov chain $\{Y_t : t \in \Gamma_n\}$ defined on a finite state space Ω with transition probability matrices M_t , $t \in \Gamma_n$, and (ii) there exists a partition $\{C_x, x = 0, 1, \dots, l\}$ on a state space Ω (where C_x and l may depend on n), such that for every $x = 0, 1, \dots, l$, $P(X_n = x) = P(Y_n \in C_x)$.

The exact distribution of a finite Markov chain imbeddable random variable can then be obtained easily from the formulation developed by Fu and Koutras (1994), which can be viewed as a special form of the Chapman–Kolmogorov theorem:

Theorem (Fu–Koutras). *If X_n can be imbedded as a finite Markov chain $\{Y_t, \Omega, M_t : t \in \Gamma_n\}$, then*

$$P(X_n = x) = P(Y_n \in C_x) = \pi_0 \left(\prod_{t=1}^n M_t \right) U(C_x), \quad (1)$$

where $\pi_0 = (P(Y_0 = a_1), P(Y_0 = a_2), \dots, P(Y_0 = a_k))$ is the initial probability of the Markov chain, $U(C_x) = \sum_{a_r \in C_x} U_r$, $U_r = (0, 0, \dots, 0, 1, 0, \dots, 0)^T$ is a $k \times 1$ unit vector with 1 at the r th coordinate and 0 elsewhere, and k is the size of the state space.

In view of the above theorem, the problem of finding the exact distribution of runs and patterns can be studied by constructing the three essential components for imbedding: (a) a proper finite state space Ω based on the structure of the specified run; (b) a proper partition $\{C_x\}$ on the state space Ω for each of the $x = 0, 1, \dots, l$; and (c) a finite Markov chain and its transition probability matrices M_t , $t = 1, \dots, n$.

2. Main result

Consider a sequence of n Bernoulli trials. Given the number of successes $N_1 = n_1$, the randomness of the sequence can be tested statistically, for example, using the Wald–Wolfowitz runs test based on the number of success runs R_n . In the past, the exact distribution of the conditional random variable, R_{n,n_1} (R_n conditional on $N_1 = n_1$), was obtained mainly from combinatorics (e.g. Wald and Wolfowitz, 1940; David, 1947). When the Bernoulli trials are not i.i.d., the combinatorial approach is very laborious (e.g. David, 1947). Alternatively, this problem can be imbedded into a finite Markov chain, and solved from a very different point of view.

Lou (1996) derived an approach for determining both the critical regions and powers of the runs test using the method of finite Markov chain imbedding, where the exact distribution of R_{n,n_1} was obtained indirectly by first finding the joint distribution of R_n and N_1 , and then projecting the distribution onto the contour $N_1 = n_1$. Under the assumption of i.i.d., the conditional distribution R_{n,n_1} can be obtained directly with the following theorem:

Theorem 1. *For a given number of successes n_1 (number of failures $n_2 = n - n_1$) in a sequence of n i.i.d. Bernoulli trials, the exact distribution of the conditional random variable R_{n,n_1} can be expressed as:*

1. for $n_1 = 0$ or 1,

$$P(R_{n,0} \equiv 0) = P(R_{n,1} \equiv 1) \equiv 1, \quad (2)$$

2. for $n_1 \geq 2$ and $1 \leq r \leq m$,

$$P(R_{n,n_1} = r) = \pi_0 \left(\prod_{t=1}^{n_2} M_t \right) U_r, \quad (3)$$

where $m = \min(n_1, n_2 + 1)$, $\pi_0 = (1, 0, \dots, 0)$ is a $1 \times m$ row vector, $U_r = (0, \dots, 0, 1, 0, \dots, 0)^T$ is an $m \times 1$ column vector with 1 at the r th coordinate and 0 elsewhere, and $M_t = (P_{ij}(t))$, $t = 1, 2, \dots, n_2$, is an $m \times m$ transition probability matrix given by

$$M_t = \begin{cases} \begin{matrix} & \begin{matrix} 1 & 2 & 3 & \dots & t & t+1 & \dots & m \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ \vdots \\ t \\ t+1 \\ \vdots \\ m \end{matrix} & \begin{bmatrix} \frac{t+1}{n_1+t} & \frac{n_1-1}{n_1+t} & & & & & \\ & \frac{t+2}{n_1+t} & \frac{n_1-2}{n_1+t} & & & & \\ & & \ddots & \ddots & & & 0 \\ & & & \ddots & \ddots & & \\ & & & & \frac{2t}{n_1+t} & \frac{n_1-t}{n_1+t} & \\ & & & & & 1 & \\ & 0 & & & & & 1 \\ & & & & & & \ddots \\ & & & & & & & 1 \end{bmatrix} \end{matrix} & 1 \leq t \leq m-1, \\ \begin{matrix} & \begin{matrix} 1 & 2 & 3 & \dots & m-1 & m \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ \vdots \\ m-1 \\ m \end{matrix} & \begin{bmatrix} \frac{t+1}{n_1+t} & \frac{n_1-1}{n_1+t} & & & & \\ & \frac{t+2}{n_1+t} & \frac{n_1-2}{n_1+t} & & 0 & \\ & & \ddots & \ddots & & \\ & & & \ddots & \ddots & \\ & 0 & & & \frac{n_1+t-1}{n_1+t} & \frac{1}{n_1+t} \\ & & & & & 1 \end{bmatrix} \end{matrix} & m \leq t \leq n_2. \end{cases} \quad (4)$$

Proof. For $n_1 = 0$ or 1, Theorem 1 is obvious. For $n_1 \geq 2$, consider a Markov chain process of randomly inserting n_2 failures F into a row of n_1 successes S , one at a time until the last F is inserted. The possible numbers of success runs are $\{1, 2, \dots, m\}$, where $m = \min(n_1, n_2 + 1)$. Define $\Omega = \{1, 2, \dots, m\}$, $\Gamma_n = \{0, 1, \dots, n_2\}$, and a Markov chain Y_t , the number of success runs after randomly inserting the t th F into a row of n_1 successes and $(t-1)$ failures, for $t \in \Gamma_n$.

The initial state prior to the insertion of failures is a sequence of n_1 successes, i.e. $Y_0 = 1$. Therefore, $P(Y_0 = 1) \equiv 1$ and $\pi_0 = (1, 0, \dots, 0)$. Suppose the first F is inserted, then $Y_1 = 2$ if and only if F is inserted between two consecutive successes. Hence,

$$P(Y_1 = 2 | Y_0 = 1) = (n_1 - 1)/(n_1 + 1),$$

and

$$P(Y_1 = 1 | Y_0 = 1) = 2/(n_1 + 1).$$

Similarly, $Y_2 = Y_1 + 1$ if and only if the second F is inserted into a space between two consecutive successes. Thus,

$$P(Y_2 = 2 | Y_1 = 1) = (n_1 - 1)/(n_1 + 2), \quad P(Y_2 = 3 | Y_1 = 2) = (n_1 - 2)/(n_1 + 2),$$

$$P(Y_2 = 1 | Y_1 = 1) = 3/(n_1 + 2), \quad P(Y_2 = 2 | Y_1 = 2) = 4/(n_1 + 2).$$

In general, when the t th F is inserted, the transition probabilities of Y_t given Y_{t-1} , $t = 1, 2, \dots, n_2$, can be expressed as

$$P_{ij}(t) = P(Y_t = j | Y_{t-1} = i) = \begin{cases} (n_1 - i)/(n_1 + t) & \text{if } j = i + 1, 1 \leq i \leq t, \\ (t + i)/(n_1 + t) & \text{if } j = i, 1 \leq i \leq t, \\ 1 & \text{if } j = i \geq t + 1, i \leq m - 1, \text{ or } j = i = m, \\ 0 & \text{otherwise.} \end{cases} \quad (4)$$

This yields the transition probability matrices M_t specified in Eq. (4), and Eq. (3) then follows immediately from Eq. (1). $\square$

Example. Let $n = 10$ and $n_1 = 4$. Then $n_2 = 6$ and $m = \min(n_1, n_2 + 1) = 4$. Consider 6 F 's inserted randomly one by one into 4 S 's. Then $\Omega = \{1, 2, 3, 4\}$, $\Gamma_n = \{0, 1, 2, 3, 4, 5, 6\}$, and the transition probability matrices are:

$$M_1 = \begin{bmatrix} \frac{2}{5} & \frac{3}{5} & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad M_2 = \begin{bmatrix} \frac{3}{6} & \frac{3}{6} & 0 & 0 \\ 0 & \frac{4}{6} & \frac{2}{6} & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad M_3 = \begin{bmatrix} \frac{4}{7} & \frac{3}{7} & 0 & 0 \\ 0 & \frac{5}{7} & \frac{2}{7} & 0 \\ 0 & 0 & \frac{6}{7} & \frac{1}{7} \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

$$M_4 = \begin{bmatrix} \frac{5}{8} & \frac{3}{8} & 0 & 0 \\ 0 & \frac{6}{8} & \frac{2}{8} & 0 \\ 0 & 0 & \frac{7}{8} & \frac{1}{8} \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad M_5 = \begin{bmatrix} \frac{6}{9} & \frac{3}{9} & 0 & 0 \\ 0 & \frac{7}{9} & \frac{2}{9} & 0 \\ 0 & 0 & \frac{8}{9} & \frac{1}{9} \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad M_6 = \begin{bmatrix} \frac{7}{10} & \frac{3}{10} & 0 & 0 \\ 0 & \frac{8}{10} & \frac{2}{10} & 0 \\ 0 & 0 & \frac{9}{10} & \frac{1}{10} \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

It follows from the above theorem that

$$P(R_{10,4} = r) = (1, 0, 0, 0) \left(\prod_{t=1}^6 M_t \right) U_r.$$

It can be seen from the above proof that finite Markov chain imbedding is not necessarily unique. Instead of inserting F 's into a row of successes, the Markov chain process can also be defined by inserting randomly n_1 successes into a row of n_2 failures, one by one. The transition probability matrices may differ, but the result is the same. To illustrate this point, re-consider the above example.

Example (continued). Now suppose instead that 4 S 's are inserted randomly one by one into 6 F 's. The process starts when the first success is inserted, and therefore $\pi_0 = (1, 0, 0, 0)$. Then $\Omega = \{1, 2, 3, 4\}$, $\Gamma_n^* = \{0, 1, 2, 3, 4\}$, and the transition probability matrices are: $M_1^* = I_{4 \times 4}$,

$$M_2^* = \begin{bmatrix} \frac{2}{8} & \frac{6}{8} & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad M_3^* = \begin{bmatrix} \frac{3}{9} & \frac{6}{9} & 0 & 0 \\ 0 & \frac{4}{9} & \frac{5}{9} & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad M_4^* = \begin{bmatrix} \frac{4}{10} & \frac{6}{10} & 0 & 0 \\ 0 & \frac{5}{10} & \frac{5}{10} & 0 \\ 0 & 0 & \frac{6}{10} & \frac{4}{10} \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Similarly,

$$P(R_{10,4} = r) = (1, 0, 0, 0) \left(\prod_{t=1}^4 M_t^* \right) U_r.$$

It is then easy to verify that, for all $r \in \{1, 2, 3, 4\}$,

$$\begin{aligned} (1, 0, 0, 0) \left(\prod_{t=1}^6 M_t \right) U_r &= (1, 0, 0, 0) \left(\prod_{t=1}^4 M_t^* \right) U_r \\ &= \left(\frac{1}{30}, \frac{9}{30}, \frac{15}{30}, \frac{5}{30} \right) U_r. \end{aligned}$$

3. Remarks

Critical regions of the test based on R_{n,n_1} have been computed in the literature for small sample sizes using the combinatory approach, and have been tabulated for reference (e.g. Swed and Eisenhart, 1943). For ease of use, results are presented here in a graphical as opposed to tabular format. Fig. 1 shows the critical regions of the two-sided test, at the 5% significance level, for a large sample of $n = 200$. Given the number of successes n_1 , the hypothesis of randomness can be tested by finding whether the observed R_{n,n_1} is located inside the critical region or not. For example, a physician may be interested in determining whether the appearance of drug side effects in a patient occurred randomly during the trial period. Suppose the patient was followed-up for 200 days, of which there are 75 days (“successes”) where side effects occurred, and the observed number of success runs r is 20. Then the null hypothesis of randomness would be rejected, since for $n_1 = 75$ and $n = 200$, $r = 20$ is located inside the critical region **Cr**.

For large samples, instead of the exact distribution of R_{n,n_1} , an approximation to the null distribution is often used. The asymptotic distribution of R_{n,n_1} , when the ratio of n_1 to $n - n_1$ remains a positive constant while both numerator and denominator approach infinity, has been proven to be normal with mean $E(R_{n,n_1}) = n_1(n - n_1 + 1)/n$ and variance $Var(R_{n,n_1}) = n_1(n_1 - 1)(n - n_1)(n - n_1 + 1)/[n^2(n - 1)]$ by Wald and Wolfowitz (1940). However, a significant discrepancy between the approximate and exact tail probabilities frequently occurs, which is due to the skewness and/or “fatness” of the distributions of runs when the ratio of number of successes to the number of trials is close to zero or one. Even when this ratio is $\frac{1}{2}$, the tail probability

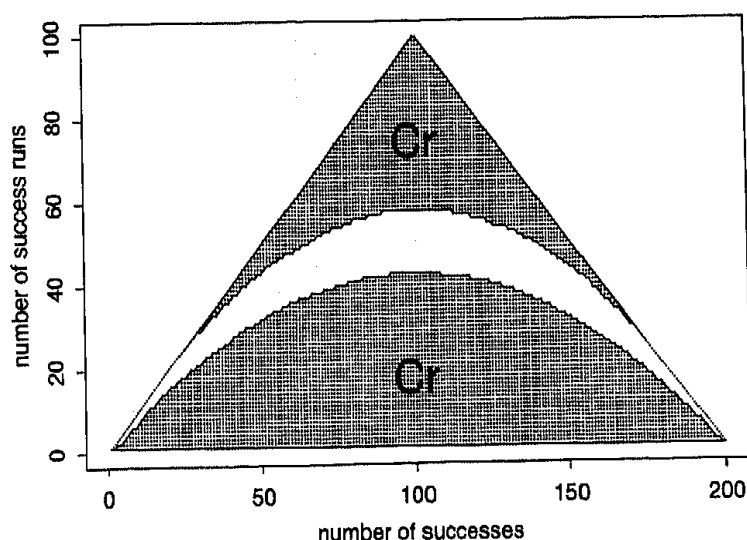


Fig. 1. Critical regions for the test based on R_{200} given the number of successes n_1 .

Table 1

The exact and approximate tail probabilities of the success runs given the number of successes for some selected cases

Tail side	Successes	Runs	Probability		
			Exact	approx. ^a	approx. ^b
$P(R_{200} \leq r)$	10	7	0.00652	0.00003	0.00069
	30	20	0.00369	0.00076	0.00193
	140	30	0.00003	0.00001	0.00002
$P(R_{200} \geq r)$	100	62	0.00003	0.00002	0.00004
	140	50	0.00990	0.00733	0.01151
	140	54	0.00007	0.00008	0.00015

^a Under the normal distribution with mean and variance as given in Section 3.

^b Normal approximation with a continuity correction term of 0.5.

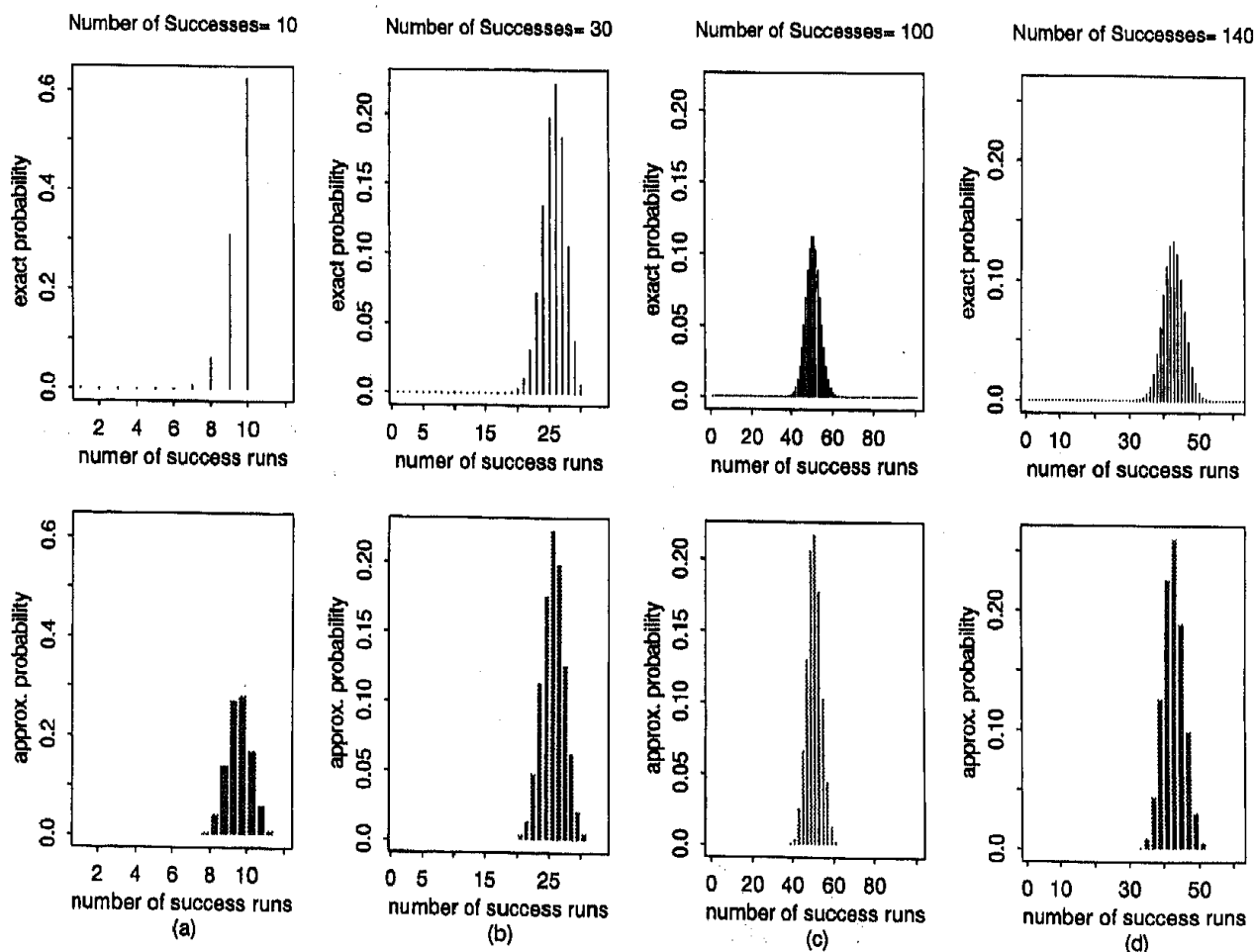


Fig. 2. (a) Top panel: the exact distribution of $R_{200,10}$. Bottom panel: histogram of 10 000 samples from $N(9.55, 0.41)$. (b) Top panel: the exact distribution of $R_{200,30}$. Bottom panel: histogram of 10 000 samples from $N(26.65, 3.18)$. (c) Top panel: the exact distribution of $R_{200,100}$. Bottom panel: histogram of 10 000 samples from $N(50.50, 12.56)$. (d) Top panel: the exact distribution of $R_{200,140}$. Bottom panel: histogram of 10 000 samples from $N(42.70, 8.95)$.

estimate from the normal approximation remains poor when the exact tail probability is very small (see Table 1). This is due to the fact that the tail probabilities of R_{n,n_1} and those of the normal distribution have different rates of convergence to zero.

To illustrate this fact, the exact and approximate probabilities are plotted in Fig. 2 for $n_1 = 10, 30, 100$, and 140, with $n = 200$. The tail probabilities for a few values of r are listed in Table 1. It is clear that even for large sample sizes (e.g. $n_1 = 140$, $n = 200$), when the observed number of success runs is away from the mean value $E(R_{n,n_1})$, the approximation is much less accurate. For example, when $R_{200,140} = 50$, the relative error of the approximate probability is 16% as opposed to 114% when $R_{200,140} = 54$, where the mean $E(R_{200,140})$ is 42.7 and the continuity correction term is 0.5. Furthermore, if the observed number of success runs is close to the critical points, approximate probabilities may lead to incorrect conclusions with regards to the null hypothesis. Therefore, the exact probabilities should always be used instead, as they are now easily available via Theorem 1.

References

- David, F.N. (1947), A power function for tests of randomness in a sequence of alternatives, *Biometrika* **34**, 335–339.
- Fu, J.C. (1986), Reliability of consecutive- k -out-of- n :F system with $(k - 1)$ step Markov dependence, *IEEE Trans. Reliability* **R35**, 602–606.
- Fu, J.C. (1996), Distribution theory of runs and patterns associated with a sequence of multi-state trials, *Statist. Sinica*, to appear.
- Fu, J.C. and M.V. Koutras (1994), Distribution theory of runs: a Markov chain approach, *J. Amer. Statist. Assoc.* **89**, 1050–1058.
- Godbole, A.P. (1990), Specific formula for some success run distributions, *Statist. Probab. Lett.* **10**, 119–124.
- Hirano, K. (1986), Some properties of the distributions of order k , in: A. N. Philippou, G. E. Bergum, and A. F. Horadam, eds. *Fibonacci Numbers and Their Applications* (Reidel, Dordrecht) pp. 43–53.
- Lou, W.Y.W. (1996), On runs and longest run tests: method of finite Markov chain imbedding, *J. Amer. Statist. Assoc.*, to appear.
- Mood, A.M. (1940), The distribution theory of runs, *Ann. Math. Statist.* **11**, 367–392.
- Swed, F.S. and C. Eisenhart (1943), Tables for testing randomness of grouping in a sequence of alternatives, *Ann. Math. Statist.* **14**, 66–87.
- Wald, A. and J. Wolfowitz (1940), On a test whether two samples are from the same population, *Ann. Math. Statist.* **11**, 147–162.